Class: - B-Tech | Branch: - CSE & CSE (DS) | Semester: - 5th
Subject: - DBMS | Code: - BTCS-501-18 | Group: A & B & DS
Duration: - 1 Hr. 30 Mins. | MM: - 24 | Time: 2:30 to 4:00 PM

Note: All Questions are Compulsory.

(Section- A)

1. Differentiate between Strong and Weak entity sets. How are they represented in an ER Diagram?
2. Define the purpose of Data Independence? Explain its various types?
3. Differentiate between Network and Relational Model?
4. Explain the difference between Delete, Drop and Truncate command of SQL? (4*2=8)

(Section- B)

5. Explain 3-tier architecture of DBMS in detail with suitable diagram?
6. Construct a ER diagram for the National Hockey Team (NHL) database using the following requirements. List your assumptions and clearly indicate the cardinality mappings as well as any role indicators in your ER diagram.

derived attribute

1. The NHL has many teams,
2. Each team has a name, a city, a coach, a captain, and a set of players,
3. Each player belongs to only one team,
4. Each player has a name, a position (such as left wing or goalie), a skill level, and a set of injury records,
5. A team captain is also a player,
6. A game is played between two teams (referred to as host team and guest team) and has a date and a score. (2*4=8)

(Section- C)

7. Create a Employee Table with following description and structure and Insert at least 50 records in the Employee Table. Assume attribute datatype and width at you own level.

Emp ID	E Name	Job	Salary	Deptt No	Deptt name	Gender	Age	DOB	Address

Write the SQL Query as per following statements.

a) Write a SQL query to fetch the number of employees working in the department 'HR'.
b) List all employee names and their salaries, whose salary lies between 20000/- and 50000/- both inclusive.
c) Write a SQL query to fetch TOP 20 records based on Salary.
d) Write a SQL query to fetch details of Employees whose employee name ends with an alphabet 'A' and contains five alphabets.
e) Write a SQL query fetch details of all employees excluding the employees name with "Sanjay" and "Sonia" from the Employee table.
f) Write a SQL query to display department where we pay 100000 as total salary to the employee in a single department excluding Job='Professor'. Also display the output in descending order of their department number.
g) Write a SQL Query to find average salary for each job if more than two employees are working in that job.
h) Write a SQL Query to modify the structure of Age column datatype as varchar2 from number datatype and width to 30. (1*8=8)

*****All the Best*****

Time allowed: 3 Hrs **Max Marks: 60**

Important Instructions:

- **All questions are compulsory**
- **Assume any missing data**

Part-A (2×10)

Q1) Answer in brief:

(a) Define Lost Update Problem with example
(b) Differentiate between Primary Key and Secondary Key
(c) What is use of Domain and Data Dependency?
(d) List the names of various DBMS and RDBMS Software's?
(e) What do you mean by integrity of data? List any two integrity constraints.
(f) Explain use of Grant and Revoke command with examples?
(g) Explain ACID properties of Transaction?
(h) What is difference between SQL and PL/SQL?
(i) Differentiate between Query processing and optimization?
(j) Define Hash Function with example?

Part-B (8×5)

Q2) What is ER Diagram? What are the symbols used in it? Design and Draw an ER diagram for a Airline Reservation system? (CO1)

OR

Discuss three-tier architecture of the database and use of mapping between schema levels? (CO1)

Q3) What are various components of relational model? How the data and relationships are converted from logical models to relational models? (CO2)

OR

Define various applications of SQL. Explain various DDL, DCL, DCL and TCL statements used in SQL along with syntax and examples. (CO2)

Q4) Define Two Phase Locking with example? How Concurrency Control plays very important role in Transaction Processing? (CO3)

OR

Define Transaction Failure. Explain various stages used in Transaction process in detail? (CO3)

Q5) Describe the term Normalization? Explain various dependences used in Normalization Also explain the difference between BCNF and 3NF briefly? (CO4)

OR

Differentiate between Backup and Recovery. Explain various Database Recovery techniques in detail? (CO4)

Q6) Define the role of Logical and Web databases. Explain the architecture of Distributed Databases Design in detail? (CO5)

OR

What is role of SQL Injection and IDS? Explain various security mechanisms used in Database systems? (CO5)

SHAHEED BHAGAT SINGH STATE UNIVERSITY, FEROZEPUR

MST-II

(Dated: -12/12/2024 Time: -10:00 to 11:30 AM)

Subject: DBMS	**Subject Code:** BTCS-501C
Branch: CSE **Semester:** 5th	**Group:** A & B
Name of the Teacher: Mr. Pawan Luthra	**Max. Marks:** -24

Note: - All Questions are compulsory. **Time:** -90 Minutes

Section-A

Q1. What is the role of IDS and SQL Injection in Database Security?

Q2 Differentiate between Serializable and Non-Serializable schedules.

Q3. Let's consider a list of numbers (34, 16, 2, 93, 80, 77,51) that has a hash table of size 10. What is the order of elements (from index 0 to size -1) in the hash table?

Q4. Differentiate between Authorization and Authentication.

Section-B

Q5. Differentiate between B Tree and B+Tree.

Q6. What is the role of Transaction? Explain its various properties and states of Transaction.

Section-C

Q7. Define Dirty Read Problem. Differentiate between Pessimistic and Optimistic Concurrency Control techniques. Explain any one Concurrency Control technique in detail.

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SHAHEED BHAGAT SINGH STATE UNIVERSITY, FEROZEPUR
B.Tech. || CSE || 5th Sem
Database Management Systems
Subject Code:BTCS 501C

Time allowed: 3 Hrs Max Marks: 60

Section A (2 marks each)

Q. 1. Short-Answer Questions: (All COs)
(a) What is an Index?
(b)What are foreign keys?
(c) Explain the concept of 'Locks' in DBMS?
(d) Explain any two Relational Algebra operations.
(e) What is the role of 'Timestamp' in DBMS?
(f) What are Triggers required in DBMS?
(g) What are clustered Indexes?
(h) What are disadvantages of using DBMS?.
(i) What is the purpose of 3rd normal form in DBMS?
(j) What is the role of a DBA?

Section B (8 marks each)

Q 2. Differentiate between DBMS and file-based system of record keeping? (CO1)

OR

Explain DDL and DML. Write two operations of each with syntax.

Q. 3. Explain 4th Normal Forms in databases with a suitable example. (CO2)

OR

How Integrity constraints are implemented in DBMS? .

Q. 4. Explain ACID properties of transactions in DBMS?

OR

Explain the concept of 'Deadlocks'? How are they resolved?

Q. 5. What are Distributed databases? In what scenarios they should be preferred?. (CO4)

OR

What are the advantages of Object Oriented Databases? Explain with a suitable example.

Q. 6. Explain different types of Recovery techniques used in DBMS. (CO5)

OR

Explain different types of 'Failure' in database management systems. How can these be handled?